

Displays

D1:

$$\mathcal{Y} = f(\mathcal{X}) = \{f(x) : x \in \mathcal{X}\}. \quad (1)$$

D2:

$$\hat{\mathcal{Y}}_\epsilon^M := \mathbf{H}(\{y_i\}_{i=1}^M) \oplus B(0, \epsilon). \quad (2)$$

D3:

$$d_{\mathbf{H}}(A, B) := \max \left(\sup_{x \in B} \inf_{y \in A} \|x - y\|, \sup_{x \in A} \inf_{y \in B} \|x - y\| \right) \quad \text{for any } A, B \in \mathcal{K}. \quad (3)$$

D4:

$$d_H(\hat{\mathcal{Y}}_{\epsilon_M}^M, \mathbf{H}(\mathcal{Y}) \oplus B(0, \bar{\epsilon})) \longrightarrow 0.$$

D5:

$$d_H(\hat{\mathcal{Y}}^M, \mathbf{H}(\mathcal{Y})) \leq \epsilon \quad \text{and} \quad \mathcal{Y} \subseteq \hat{\mathcal{Y}}_\epsilon^M.$$

D6:

$$d_H(\hat{\mathcal{Y}}^M, \mathbf{H}(\mathcal{Y})) \leq \epsilon \quad \text{and} \quad \mathcal{Y} \subseteq \hat{\mathcal{Y}}_\epsilon^M.$$

D7:

$$\min_{(\mu, \nu)} \sum_{t=1}^N (\mu_t - x_{\text{goal}})^\top Q(\mu_t - x_{\text{goal}}) + \sum_{t=1}^N \nu_t^\top R \nu_t, \quad \text{s.t.} \quad \mu_0 = x_0, \quad (4a)$$

D8:

$$\mu_{t+1} = f(\mu_t, \nu_t, \bar{m}, \bar{F}), \quad \nu_t \in \mathcal{U}, \quad \mathcal{X}_t(\nu) \subset \mathcal{X}_{\text{free}}, \quad \mathcal{X}_N(\nu) \subset \mathcal{X}_{\text{goal}}, \quad t = 0, \dots, N-1. \quad (4b)$$

D9:

$$d_{\mathbf{H}}(A, B) = \max \left(\sup_{x \in B} \inf_{y \in A} \|x - y\|, \sup_{x \in A} \inf_{y \in B} \|x - y\| \right). \quad (5)$$

D10:

$$\mathbb{P}(\hat{\mathcal{Y}}^M \cap K \neq \emptyset \text{ infinitely often}) = \mathbb{P} \left(\bigcap_{N=1}^{\infty} \bigcup_{M=N}^{\infty} \{\hat{\mathcal{Y}}^M \cap K \neq \emptyset\} \right) = 0. \quad (C1)$$

D11:

$$\mathbb{P}(\hat{\mathcal{Y}}^M \cap G = \emptyset \text{ infinitely often}) = \mathbb{P} \left(\bigcap_{N=1}^{\infty} \bigcup_{M=N}^{\infty} \{\hat{\mathcal{Y}}^M \cap G = \emptyset\} \right) = 0. \quad (C2)$$

D12:

$$d_H(\hat{\mathcal{Y}}_{\epsilon_M}^M, \mathbf{H}(\mathcal{Y}) \oplus B(0, \bar{\epsilon})) \longrightarrow 0.$$

D13:

$$\mathbb{P} \left(\bigcup_{M=0}^{\infty} (y_{2M+1} \in (G_{\partial}^1 \oplus B(0, \bar{\epsilon}))) \cap (y_{2M+2} \in (G_{\partial}^2 \oplus B(0, \bar{\epsilon}))) \right) = 1. \quad (6)$$

D14:

$$\mathbb{P}(\hat{\mathcal{Y}}_{\epsilon_M}^M \cap G = \emptyset \text{ i.o.}) = \mathbb{P} \left(\bigcap_{N=1}^{\infty} \bigcup_{M=N}^{\infty} \hat{\mathcal{Y}}_{\epsilon_M}^M \cap G = \emptyset \right) = 1 - \mathbb{P} \left(\bigcup_{N=1}^{\infty} \bigcap_{M=N}^{\infty} \hat{\mathcal{Y}}_{\epsilon_M}^M \cap G \neq \emptyset \right).$$

D15:

$$\mathbb{P} \left(\bigcup_{N=1}^{\infty} \bigcap_{M=N}^{\infty} \hat{\mathcal{Y}}_{\epsilon_M}^M \cap G \neq \emptyset \right) \geq \mathbb{P} \left(\bigcup_{N=M_{\epsilon}}^{\infty} \bigcap_{M=N}^{\infty} \hat{\mathcal{Y}}_{\epsilon_M}^M \cap G \neq \emptyset \right) \geq \mathbb{P} \left(\bigcup_{N=M_{\epsilon}}^{\infty} \bigcap_{M=N}^{\infty} \hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M \cap G \neq \emptyset \right).$$

D16:

$$\mathbb{P} \left(\bigcup_{N=M_{\epsilon}}^{\infty} \bigcap_{M=N}^{\infty} \hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M \cap G \neq \emptyset \right) = \mathbb{P} \left(\bigcup_{M=M_{\epsilon}}^{\infty} \hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M \cap G \neq \emptyset \right).$$

D17:

$$\mathbb{P} \left(\bigcup_{M=M_{\epsilon}}^{\infty} \hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M \cap G \neq \emptyset \right) = 1 \implies \mathbb{P}(\hat{\mathcal{Y}}_{\epsilon_M}^M \cap G = \emptyset \text{ i.o.}) = 0. \quad (7)$$

D18:

$$\begin{aligned} & \{\omega \in \Omega : (y_{2M+1}(\omega) \in (G_\partial^1 \oplus B(0, \bar{\epsilon}))) \cap (y_{2M+2}(\omega) \in (G_\partial^2 \oplus B(0, \bar{\epsilon})))\} \\ & \subseteq \{\omega \in \Omega : (\hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^{2M+2}(\omega) \cap (G_\partial^1 \oplus B(0, \bar{\epsilon})) \neq \emptyset) \cap (\hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^{2M+2}(\omega) \cap (G_\partial^2 \oplus B(0, \bar{\epsilon})) \neq \emptyset)\} \end{aligned}$$

D19:

$$\begin{aligned} & \bigcup_{M=0}^{\infty} \{\omega \in \Omega : (y_{2M+1}(\omega) \in (G_\partial^1 \oplus B(0, \bar{\epsilon}))) \cap (y_{2M+2}(\omega) \in (G_\partial^2 \oplus B(0, \bar{\epsilon})))\} \\ & \subseteq \bigcup_{M=2}^{\infty} \{\omega \in \Omega : (\hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M(\omega) \cap G_\partial^1 \neq \emptyset) \cap (\hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M(\omega) \cap G_\partial^2 \neq \emptyset)\} \\ & \subseteq \bigcup_{M=1}^{\infty} \{\omega \in \Omega : (\hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M(\omega) \cap G_\partial^1 \neq \emptyset) \cap (\hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M(\omega) \cap G_\partial^2 \neq \emptyset)\}. \end{aligned}$$

D20:

$$\mathbb{P} \left(\bigcup_{M=M_\epsilon}^{\infty} (\hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M \cap G_\partial^1 \neq \emptyset) \cap (\hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M \cap G_\partial^2 \neq \emptyset) \right) = 1.$$

D21:

$$\mathbb{P} \left(\bigcup_{M=M_\epsilon}^{\infty} \hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M \cap G \neq \emptyset \right) \geq \mathbb{P} \left(\bigcup_{M=M_\epsilon}^{\infty} (\hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M \cap G_\partial^1 \neq \emptyset) \cap (\hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M \cap G_\partial^2 \neq \emptyset) \right) = 1.$$

D22:

$$D(\partial\mathcal{Y}, \epsilon) \leq D(\partial\mathcal{X}, \epsilon/L).$$

D23:

$$\mathbb{P}_{\mathcal{Y}}(B(y, \delta)) \geq \mathbb{P}_{\mathcal{X}}(B(x, \epsilon)) \quad \text{for all } \epsilon \in [0, \delta/L].$$

D24:

$$Y_\epsilon^M = \bigcup_{i=1}^M B(y_i, \epsilon), \quad \pi(\partial\mathcal{Y}, Y_\epsilon^M) = \sup_{y \in \partial\mathcal{Y}} \mathbb{P}(\{y\} \cap Y_\epsilon^M = \emptyset).$$

D25:

$$\mathbb{P}(\partial\mathcal{Y} \subset Y_{2\epsilon}^M) \geq 1 - D(\partial\mathcal{Y}, \epsilon)\pi(\partial\mathcal{Y}, Y_\epsilon^M).$$

D26:

$$f(B(x, \epsilon)) \subseteq B(y, \delta) \quad \forall \epsilon \in [0, \delta/L], \tag{8}$$

D27:

$$\sup_{y_i \in F_{\partial\mathcal{Y}}} \|y - y_i\| = \sup_{x_i \in F_{\partial\mathcal{X}}} \|f(x) - f(x_i)\| \leq \sup_{x_i \in F_{\partial\mathcal{X}}} L\|x - x_i\| \leq \epsilon.$$

D28:

$$\mathbb{P}_{\mathcal{Y}}(B(y, \delta)) = \mathbb{P}_{\mathcal{X}}(f^{-1}(B(y, \delta))) \geq \mathbb{P}_{\mathcal{X}}(B(x, \epsilon)) \quad \forall \epsilon \in [0, \delta/L],$$

D29:

$$\begin{aligned} \mathbb{P}(\partial\mathcal{Y} \not\subset Y_{2\epsilon}^M) & \leq \mathbb{P}(B(F_{\partial\mathcal{Y}}, \epsilon) \not\subset Y_{2\epsilon}^M) = \mathbb{P}(F_{\partial\mathcal{Y}} \not\subset Y_\epsilon^M) \\ & = \mathbb{P} \left(\bigcup_{y_i \in F_{\partial\mathcal{Y}}} y_i \notin Y_\epsilon^M \right) \\ & \leq \sum_{y_i \in F_{\partial\mathcal{Y}}} \mathbb{P}(y_i \notin Y_\epsilon^M) = \sum_{y_i \in F_{\partial\mathcal{Y}}} \mathbb{P}(\{y_i\} \cap Y_\epsilon^M = \emptyset) \\ & \leq |F_{\partial\mathcal{Y}}| \cdot \sup_{y \in \partial\mathcal{Y}} \mathbb{P}(\{y\} \cap Y_\epsilon^M = \emptyset) \\ & \leq D(\partial\mathcal{Y}, \epsilon)\pi(\partial\mathcal{Y}, Y_\epsilon^M). \end{aligned}$$

D30:

$$\mathbb{P}(d_H(\hat{\mathcal{Y}}^M, \mathcal{H}(\mathcal{Y})) \leq \epsilon) \geq 1 - \delta_M \quad \text{and} \quad \mathbb{P}(\mathcal{Y} \subseteq \hat{\mathcal{Y}}_\epsilon^M) \geq 1 - \delta_M.$$

D31:

$$\pi(\partial\mathcal{Y}, Y_\epsilon^M) = \sup_{y \in \partial\mathcal{Y}} \mathbb{P}(\{y\} \cap Y_\epsilon^M = \emptyset) = \sup_{y \in \partial\mathcal{Y}} \mathbb{P}(B(y, \epsilon) \cap Y^M = \emptyset),$$

D32:

$$\begin{aligned}
\pi(\partial\mathcal{Y}, Y_\epsilon^M) &= \sup_{y \in \partial\mathcal{Y}} \mathbb{P}(B(y, \epsilon) \cap Y^M = \emptyset) \\
&= \sup_{y \in \partial\mathcal{Y}} \mathbb{P}\left(\bigcap_{i=1}^M (y_i \notin B(y, \epsilon))\right) \\
&= \left(1 - \inf_{y \in \partial\mathcal{Y}} \mathbb{P}(y_i \in B(y, \epsilon))\right)^M \\
&= \left(1 - \inf_{y \in \partial\mathcal{Y}} \mathbb{P}_{\mathcal{Y}}(B(y, \epsilon))\right)^M.
\end{aligned}$$

D33:

$$\pi(\partial\mathcal{Y}, Y_\epsilon^M) \leq \left(1 - \inf_{x \in \partial\mathcal{X}} \mathbb{P}_{\mathcal{X}}(B(x, \epsilon/L))\right)^M.$$

D34:

$$\pi(\partial\mathcal{Y}, Y_{\epsilon/2}^M) \leq \left(1 - \inf_{x \in \partial\mathcal{X}} \mathbb{P}_{\mathcal{X}}(B(x, \epsilon/(2L)))\right)^M \leq (1 - \Lambda_\epsilon^L)^M.$$

D35:

$$\mathbb{P}(\partial\mathcal{Y} \subseteq Y_\epsilon^M) \geq 1 - D(\partial\mathcal{Y}, \epsilon/2)\pi(\partial\mathcal{Y}, Y_{\epsilon/2}^M).$$

D36:

$$\begin{aligned}
\mathbb{P}(d_H(\hat{\mathcal{Y}}^M, \mathcal{H}(\mathcal{Y})) \leq \epsilon) &\geq \mathbb{P}(\partial\mathcal{Y} \subseteq Y_\epsilon^M) \\
&\geq 1 - D(\partial\mathcal{Y}, \epsilon/2)\pi(\partial\mathcal{Y}, Y_{\epsilon/2}^M) \\
&\geq 1 - D(\partial\mathcal{X}, \epsilon/(2L))(1 - \Lambda_\epsilon^L)^M.
\end{aligned}$$

D37:

$$\inf_{x \in \partial\mathcal{X}} \lambda(\mathcal{X} \cap B(x, \epsilon)) \geq \lambda(B(0, \epsilon) \cap B(\mathbf{r}, r)).$$

D38:

$$\lambda(\mathcal{X} \cap B(x, \epsilon)) \geq \lambda(B(\tilde{x}, r) \cap B(x, \epsilon)) = \lambda(B(x - \tilde{x}, r) \cap B(0, \epsilon)) = \lambda(B(\mathbf{r}, r) \cap B(0, \epsilon))$$

D39:

$$\mathbb{P}(d_H(\hat{\mathcal{Y}}^M, \mathcal{H}(\mathcal{Y})) \leq \epsilon) \geq 1 - \delta_M, \quad \text{and} \quad \mathbb{P}(\mathcal{Y} \subseteq \hat{\mathcal{Y}}_\epsilon^M) \geq 1 - \delta_M.$$

D40:

$$V(r, a) = \begin{cases} \frac{\pi^{p/2}}{2\Gamma(\frac{p}{2}+1)} r^n I_{1-a^2/r^2}(\frac{n+1}{2}, \frac{1}{2}) & \text{if } a \geq 0, \\ \frac{\pi^{p/2}}{2\Gamma(\frac{p}{2}+1)} r^n (2 - I_{1-a^2/r^2}(\frac{n+1}{2}, \frac{1}{2})) & \text{otherwise.} \end{cases}$$

D41:

$$f(x) = W^\ell x^\ell + b^\ell, \quad x^{k+1} = \phi^k(W^k x^k + b^k), \quad k = 0, \dots, \ell - 1, \quad x^0 = x,$$

D42:

$$\|f(x_1) - f(x_2)\| \leq L\|x_1 - x_2\| \quad \forall x_1, x_2 \in \mathcal{X}.$$

D43:

$$L = \max_{i=1, \dots, N} \{\|\nabla f(x_i)\| \text{ for some } x_i \in A_i\}.$$

D44:

$$\mathbb{P}(f \text{ is } \hat{L}\text{-Lipschitz continuous over } \mathcal{X}) \geq 1 - \delta_M.$$

D45:

$$\begin{aligned}
\mathbb{P}\left(\bigcap_{i=1}^N \left\{\bigcup_{j=1}^M \{x_j\} \cap A_i \neq \emptyset\right\}\right) &= 1 - \mathbb{P}\left(\bigcup_{i=1}^N \left\{\bigcup_{j=1}^M \{x_j\} \cap A_i \neq \emptyset\right\}^c\right) \\
&\geq 1 - \sum_{i=1}^N \mathbb{P}\left(\bigcap_{j=1}^M x_j \notin A_i\right) \quad (\text{Boole's inequality}) \\
&= 1 - \sum_{i=1}^N \mathbb{P}(x_j \notin A_i)^M \quad (\text{independent samples}) \\
&\geq 1 - N(1 - \Lambda_N)^M,
\end{aligned}$$

Inline

- I1: 1
- I2: 2
- I3: 3
- I4: ϵ
- I5: M
- I6: x_i
- I7: $\mathcal{X}$
- I8: f
- I9: $\hat{\mathcal{Y}}_\epsilon^M$
- I10: $\mathcal{Y}$
- I11: r
- I12: $\lambda(\cdot)$
- I13: $\mathbb{R}^p$
- I14: $\Gamma(\cdot)$
- I15: $\mathrm{H}(A)$
- I16: $A \subset \mathbb{R}^n$
- I17: $A^c = \mathbb{R}^n \setminus A$
- I18: ∂A
- I19: $\oplus$
- I20: $B(x, r) := \{y \in \mathbb{R}^n : \|y - x\| \leq r\}$
- I21: $x \in \mathbb{R}^n$
- I22: $r \geq 0$
- I23: $\mathring{B}(x, r)$
- I24: $\mathbb{R}^n$
- I25: $\mathcal{K}$
- I26: $A \in \mathcal{K}$
- I27: $d > 0$
- I28: $D(A, d) := \min\{n \in \mathbb{N} : \exists \{a_1, \dots, a_n\} \subset \mathbb{R}^n, A \subset B(a_1, d) \cup \dots \cup B(a_n, d)\}$
- I29: d
- I30: A
- I31: $\mathcal{X} \subset \mathbb{R}^p$
- I32: $f : \mathbb{R}^p \rightarrow \mathbb{R}^n$
- I33: $y = f(x)$
- I34: $x \in \mathcal{X}$
- I35: $\mathcal{Y} \subset \mathbb{R}^n$
- I36: $M \in \mathbb{N}$
- I37: $\epsilon > 0$
- I38: $\mathbb{P}_{\mathcal{X}}$
- I39: $\mathbb{R}^p$
- I40: $y_i = f(x_i)$
- I41: y_i
- I42: σ
- I43: $\mathrm{H}(\mathcal{Y})$
- I44: $\mathbb{P}_{\mathcal{X}}(\mathcal{X}) = 1$
- I45: $\mathbb{P}_{\mathcal{X}}(\{x \in \mathcal{X} : f(x) \in \mathring{B}(y, r)\}) > 0$
- I46: $y \in \partial \mathcal{Y}$
- I47: $r > 0$
- I48: $\mathbb{P}_{\mathcal{X}}(f^{-1}(A)) > 0$
- I49: $\mathcal{Y} \cap A \neq \emptyset$
- I50: $\partial \mathcal{Y}$
- I51: $\bar{\epsilon} \geq 0$
- I52: $(\epsilon_M)_{M \in \mathbb{N}}$
- I53: $\epsilon_M \geq 0$
- I54: $\epsilon_M \rightarrow \bar{\epsilon}$
- I55: $M \rightarrow \infty$

I56: $\epsilon \geq 0$
 I57: $\hat{\mathcal{Y}}_\epsilon^M = \mathcal{H}(\{y_i\}_{i=1}^M) \oplus B(0, \epsilon)$
 I58: $\blacksquare$
 I59: $\epsilon_M = 1/M$
 I60: $\hat{\mathcal{Y}}_{\epsilon_M}^M$
 I61: $\bar{\epsilon}$
 I62: $\mathcal{Y}_\epsilon^M$
 I63: $\mathcal{H}(\mathcal{Y}) \oplus B(0, \epsilon)$
 I64: $\epsilon_M = \bar{\epsilon} = 0$
 I65: ϵ_M
 I66: L
 I67: $L \geq 0$
 I68: $\|f(x_1) - f(x_2)\| \leq L \|x_1 - x_2\|$
 I69: $x_1, x_2 \in \mathcal{X}$
 I70: $\partial\mathcal{X}$
 I71: $\epsilon, L > 0$
 I72: $\Lambda_\epsilon^L > 0$
 I73: $\mathbb{P}_\mathcal{X}(B(x, \frac{\epsilon}{2L})) \geq \Lambda_\epsilon^L$
 I74: $x \in \partial\mathcal{X}$
 I75: Λ_ϵ^L
 I76: $\epsilon/(2L)$
 I77: x
 I78: $D(\partial\mathcal{X}, d)$
 I79: $\hat{\mathcal{Y}}^M = \mathcal{H}(\{y_i\}_{i=1}^M)$
 I80: $\delta_M = D(\partial\mathcal{X}, \epsilon/(2L))(1 - \Lambda_\epsilon^L)^M$
 I81: $\partial\mathcal{Y} \subseteq f(\partial\mathcal{X})$
 I82: $1 - \delta_M$
 I83: $D(\mathcal{X}, \epsilon)$
 I84: $D(\partial\mathcal{X}, \epsilon) \leq D(\mathcal{X}, \epsilon)$
 I85: $\mathcal{X}^c$
 I86: 2
 I87: $\tilde{x} \in \mathcal{X}$
 I88: $x \in B(\tilde{x}, r) \subseteq \mathcal{X}$
 I89: $\mathcal{X} \oplus B(0, r)$
 I90: $\mathbb{P}_\mathcal{X}(A) \geq p_0 \lambda(A)$
 I91: $A \subset \mathcal{X}$
 I92: $p_0 > 0$
 I93: $p_\mathcal{X} : \mathbb{R}^p \rightarrow \mathbb{R}_+$
 I94: $\mathbb{P}_\mathcal{X}(A) = \int_A p_\mathcal{X}(x) dx \geq p_0 \int_A dx = p_0 \lambda(A)$
 I95: $\mathbf{r} = (r, 0, \dots, 0) \in \mathbb{R}^p$
 I96: $\Lambda_\epsilon^{r,L} = \lambda(B(0, \epsilon/(2L)) \cap B(\mathbf{r}, r))$
 I97: $\delta_M = D(\partial\mathcal{X}, \epsilon/(2L))(1 - p_0 \Lambda_\epsilon^{r,L})^M$
 I98: $\Lambda_\epsilon^L = p_0 \Lambda_\epsilon^{r,L}$
 I99: $\Lambda_\epsilon^{r,L}$
 I100: p
 I101: $\mathbb{P}_\mathcal{X}$
 I102: $y_i = f(x_i) \in \mathcal{Y}$
 I103: $\mathcal{X} \in \mathcal{K}$
 I104: $d = \sup_{x \in \partial\mathcal{X}} \|x\|$
 I105: $D(\partial\mathcal{X}, \epsilon) \leq (2d\sqrt{n}/\epsilon)^n$
 I106: $\mathcal{X} = B(0, 1)$
 I107: $f(x) = (Lx_1, x_2)$
 I108: $L \geq 1$
 I109: 1
 I110: $\mathbb{P}_\mathcal{X}^\alpha$
 I111: $\alpha \geq 1$

I112: $\alpha = 1$
 I113: $\alpha \rightarrow \infty$
 I114: $\delta_M = 10^{-3}$
 I115: $\mathbb{P}(d_H(\hat{\mathcal{Y}}^M, \mathcal{Y}) \leq \epsilon) \geq 1 - \delta_M$
 I116: $M = 1000$
 I117: α
 I118: $N = 9$
 I119: $\mathcal{X}_0$
 I120: 16
 I121: $M = 10^3$
 I122: $\epsilon = 0.02$
 I123: $u_t = \pi_{\text{nn}}(x_t)$
 I124: $x_{t+1} = Ax_t + Bu_t$
 I125: $t \in \mathbb{N}$
 I126: $x_t \in \mathbb{R}^2$
 I127: $u_t \in \mathbb{R}$
 I128: $\mathcal{X}_0 \subset \mathbb{R}^2$
 I129: $\mathcal{X}_t = \{(A(\cdot) + B\pi_{\text{nn}}(\cdot)) \circ \dots \circ (Ax_0 + B\pi_{\text{nn}}(x_0)) : x_0 \in \mathcal{X}_0\}$
 I130: $(\mathcal{X}, \mathcal{Y}) = (\mathcal{X}_0, \mathcal{X}_t)$
 I131: $f(x) = (A(\cdot) + B\pi_{\text{nn}}(\cdot)) \circ \dots \circ (Ax + B\pi_{\text{nn}}(x))$
 I132: π_{nn}
 I133: 5
 I134: ¹
 I135: $\epsilon = 0$
 I136: $M = 10^6$
 I137: 100
 I138: $\mathcal{Y} = \mathcal{X}_4$
 I139: ²
 I140: 1400
 I141: $1 - 10^{-4}$
 I142: $L = 1$
 I143: 9
 I144: $t \geq 0$
 I145: $x_t \in \mathbb{R}^6$
 I146: $u_t \in \mathbb{R}^3$
 I147: $x_{t+1} = f(x_t, u_t, m, F)$
 I148: $m \in [10, 18] \text{ kg}$
 I149: $F = (F_x, F_y) \in [-0.015, 0.015]^2 \text{ N}$
 I150: $x_0 \in \mathbb{R}^n$
 I151: $\mathcal{X}_{\text{goal}} \subset \mathbb{R}^n$
 I152: $\mathcal{X}_{\text{free}}$
 I153: $\mu = (\mu_0, \dots, \mu_N)$
 I154: $\nu = (\nu_0, \dots, \nu_{N-1})$
 I155: $(\bar{m}, \bar{F}_x, \bar{F}_y) = (14, 0, 0)$
 I156: $x_{\text{goal}} \in \mathcal{X}_{\text{goal}}$
 I157: $\mathcal{X}_t(\nu) \subset \mathbb{R}^n$
 I158: $\mathcal{X}_t(\nu) = \{x_t = f(\cdot, \nu_{t-1}, m, F) \circ \dots \circ f(x_0, \nu_0, m, F) : (m, F) \in [10, 18] \times [-0.015, 0.015]\}$
 I159: $\epsilon = 0.03$
 I160: 10
 I161: $(m, F) \in \{14\} \times \{(0, 0)\}$
 I162: δ
 I163: $\hat{\mathcal{Y}}_\epsilon^M$
 I164: $\mathcal{B}(\mathbb{R}^n)$
 I165: $\|\cdot\|$
 I166: $B(x, r) = \{y \in \mathbb{R}^n : \|y - x\| \leq r\}$
 I167: $A, B \in \mathcal{K}$

I168: $\mathcal{B}(\mathcal{K})$
 I169: $(\mathcal{K}, \mathcal{B}(\mathcal{K}))$
 I170: $(\Omega, \mathcal{G}, \mathbb{P})$
 I171: $\hat{\mathcal{Y}} : \Omega \rightarrow \mathcal{K}$
 I172: $\{\omega \in \Omega : \hat{\mathcal{Y}}(\omega) \in \mathcal{Y}\} \in \mathcal{G}$
 I173: $\mathcal{Y} \in \mathcal{B}(\mathcal{K})$
 I174: $\hat{\mathcal{Y}}$
 I175: $\mathbb{P}(\hat{\mathcal{Y}} \in \mathcal{Y})$
 I176: $T_{\hat{\mathcal{Y}}} : \mathcal{K} \rightarrow [0, 1]$
 I177: $K \in \mathcal{K}$
 I178: $T_{\hat{\mathcal{Y}}}(K) = \mathbb{P}(\hat{\mathcal{Y}} \cap K \neq \emptyset)$
 I179: K
 I180: $\mathcal{Y} \in \mathcal{K}$
 I181: $\{\hat{\mathcal{Y}}^M\}_{M=1}^{\infty}$
 I182: $\mathcal{Y} \cap K = \emptyset$
 I183: $G \subset \mathbb{R}^n$
 I184: $\mathcal{Y} \cap G \neq \emptyset$
 I185: $\{\mathcal{Y}^M\}_{M=1}^{\infty}$
 I186: $d_H(\mathcal{Y}^M, \mathcal{Y}) \rightarrow 0$
 I187: $(\mathbb{R}^p, \mathcal{B}(\mathbb{R}^p))$
 I188: $\hat{\mathcal{Y}}_{\epsilon}^M = \mathbf{H}(\{y_i\}_{i=1}^M) \oplus B(0, \epsilon)$
 I189: $\mathcal{G}$
 I190: $\mathbb{P}_{\mathcal{X}}(A) = \mathbb{P}(x_i \in A)$
 I191: $A \in \mathcal{B}(\mathbb{R}^p)$
 I192: 3
 I193: $\mathbb{P}_{\mathcal{Y}}$
 I194: $\mathbb{P}_{\mathcal{Y}}(B) = \mathbb{P}(y_i \in B) = \mathbb{P}_{\mathcal{X}}(f^{-1}(B))$
 I195: $B \in \mathcal{B}(\mathbb{R}^n)$
 I196: $\hat{\mathcal{Y}}_{\epsilon}^M : \Omega \rightarrow \mathcal{K}$
 I197: 4
 I198: $x_i(\omega)$
 I199: $y_i(\omega)$
 I200: $\hat{\mathcal{Y}}_{\epsilon}^M(\omega) \in \mathcal{K}$
 I201: $\omega \in \Omega$
 I202: $\Omega = \mathbb{R}^p \times \cdots \times \mathbb{R}^p$
 I203: $\mathcal{G} = \mathcal{B}(\mathbb{R}^p) \otimes \cdots \otimes \mathcal{B}(\mathbb{R}^p)$
 I204: $\mathbb{P} = \mathbb{P}_{\mathcal{X}} \otimes \cdots \otimes \mathbb{P}_{\mathcal{X}}$
 I205: $x = (x_1, \dots, x_M) : \Omega \rightarrow \Omega : \omega \mapsto \omega$
 I206: $\hat{\mathcal{Y}}_{\epsilon}^M(\omega)$
 I207: $\mathbb{P}$
 I208: $\hat{\mathcal{Y}}^M = \mathbf{H}(\{y_i\}_{i=1}^M)$
 I209: $\hat{\mathcal{Y}}_{\epsilon_M}^M = \mathbf{H}(\{y_i\}_{i=1}^M) \oplus B(0, \epsilon_M)$
 I210: $\{\hat{\mathcal{Y}}_{\epsilon_M}^M(\omega), M \geq 1\}$
 I211: $\mathbf{H}(\mathcal{Y}) \oplus B(0, \bar{\epsilon})$
 I212: $K \in \mathcal{K}$
 I213: $(\mathbf{H}(\mathcal{Y}) \oplus B(0, \bar{\epsilon})) \cap K = \emptyset$
 I214: $(\mathbf{H}(\mathcal{Y}) \oplus B(0, \bar{\epsilon} + \epsilon)) \cap K = \emptyset$
 I215: 5
 I216: $\epsilon_M \rightarrow \bar{\epsilon}$
 I217: $\bar{\epsilon} - \epsilon < \epsilon_M < \bar{\epsilon} + \epsilon$
 I218: $M \geq M_{\epsilon}$
 I219: $y_i(\omega) \in \mathcal{Y}$
 I220: i
 I221: $\hat{\mathcal{Y}}^M \subseteq \mathbf{H}(\mathcal{Y})$
 I222: $(\hat{\mathcal{Y}}^M \oplus B(0, \epsilon_M)) \subseteq (\mathbf{H}(\mathcal{Y}) \oplus B(0, \epsilon_M)) \subset (\mathbf{H}(\mathcal{Y}) \oplus B(0, \bar{\epsilon} + \epsilon))$
 I223: $A \cap B = \emptyset$

I224: $A \cap (B \oplus B(0, \epsilon)) = \emptyset$
 I225: $(\hat{\mathcal{Y}}^M \oplus B(0, \epsilon_M)) \cap K = \emptyset$
 I226: $\mathbb{P}(\hat{\mathcal{Y}}^M \oplus B(0, \epsilon_M) \cap K \neq \emptyset) = 0$
 I227: $\sum_{M=1}^{\infty} \mathbb{P}(\hat{\mathcal{Y}}^M \oplus B(0, \epsilon_M) \cap K \neq \emptyset) < \infty$
 I228: $\mathbb{P}(\hat{\mathcal{Y}}^M \oplus B(0, \epsilon_M) \cap K \neq \emptyset \text{ i.o.}) = 0$
 I229: $(H(\mathcal{Y}) \oplus B(0, \bar{\epsilon})) \cap G \neq \emptyset$
 I230: $H(\mathcal{Y}) \cap (G \oplus B(0, \bar{\epsilon})) \neq \emptyset$
 I231: $\mathbb{P}(\hat{\mathcal{Y}}_{\epsilon_M}^M \cap G = \emptyset \text{ i.o.}) = 0$
 I232: $G_{\partial}^1, G_{\partial}^2 \subset \mathbb{R}^n$
 I233: $\partial \mathcal{Y} \cap (G_{\partial}^1 \oplus B(0, \bar{\epsilon})) \neq \emptyset$
 I234: $\partial \mathcal{Y} \cap (G_{\partial}^2 \oplus B(0, \bar{\epsilon})) \neq \emptyset$
 I235: G_{∂}^j
 I236: G
 I237: $\bar{\epsilon} = 0$
 I238: $H(\mathcal{Y}) \cap G \neq \emptyset$
 I239: $\partial H(\mathcal{Y}) \cap G_{\partial}^j \neq \emptyset$
 I240: $y_1 \in G_{\partial}^1$
 I241: $y_2 \in G_{\partial}^2$
 I242: $H(\{y_1, y_2\}) \cap G \neq \emptyset$
 I243: $\hat{\mathcal{Y}}^M$
 I244: G_{∂}^1
 I245: G_{∂}^2
 I246: $\hat{\mathcal{Y}}_{\epsilon}^M \subseteq \hat{\mathcal{Y}}_{\epsilon}^{M+1}$
 I247: $\hat{\mathcal{Y}}_{\epsilon_M}^M$
 I248: $G \oplus B(0, \bar{\epsilon})$
 I249: $A_{2i} = \{\omega \in \Omega \mid y_{2i+1} \in (G_{\partial}^1 \oplus B(0, \bar{\epsilon})), y_{2(i+1)} \in (G_{\partial}^2 \oplus B(0, \bar{\epsilon}))\}$
 I250: $i \in \mathbb{N}$
 I251: $\{A_{2i}\}_{i=0}^{\infty}$
 I252: $\mathbb{P}(A_{2i}) = \mathbb{P}(y_{2i+1} \in (G_{\partial}^1 \oplus B(0, \bar{\epsilon})))\mathbb{P}(y_{2i+2} \in (G_{\partial}^2 \oplus B(0, \bar{\epsilon})))$
 I253: $j = 1, 2$
 I254: $\mathbb{P}(A_{2i}) \geq \delta$
 I255: $\delta > 0$
 I256: $\bar{\epsilon}$
 I257: $\sum_{i=0}^{\infty} \mathbb{P}(A_{2i}) = \infty$
 I258: $\mathbb{P}(\bigcup_{N=0}^{\infty} \bigcap_{M=N}^{\infty} A_{2M}) = 1$
 I259: $\bigcap_{N=0}^{\infty} A_n \subseteq A_0$
 I260: A_i
 I261: $\mathbb{P}(\bigcup_{N=0}^{\infty} \bigcap_{M=N}^{\infty} A_{2M}) \leq \mathbb{P}(\bigcup_{M=0}^{\infty} A_{2M})$
 I262: $H(\mathcal{Y}) \cap (G \oplus B(0, \bar{\epsilon})) \neq \emptyset$
 I263: $H(\mathcal{Y}) \cap (G \oplus B(0, \bar{\epsilon} - \epsilon)) \neq \emptyset$
 I264: 6
 I265: $\epsilon \rightarrow \bar{\epsilon}$
 I266: $M_{\epsilon} \in \mathbb{N}$
 I267: $M \geq M_{\epsilon}$
 I268: $\hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M \subset \hat{\mathcal{Y}}_{\epsilon_M}^M$
 I269: $\hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M \subseteq \hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^{M+1}$
 I270: $\{\omega \in \Omega \mid \bigcap_{M=N}^{\infty} \hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M \cap G \neq \emptyset\} = \{\omega \in \Omega \mid \hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^N \cap G \neq \emptyset\}$
 I271: $M \geq 0$
 I272: $A \subset \mathcal{K}$
 I273: $B \subset \mathbb{R}^n$
 I274: $A \cap B \neq \emptyset \implies A \cap (B \oplus B(0, \epsilon)) \neq \emptyset$
 I275: $G_{\delta}^1, G_{\delta}^2$
 I276: 7
 I277: $\mathbb{P}\left(\bigcup_{M=M_{\epsilon}}^{\infty} \hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M \cap G \neq \emptyset\right) = 1$
 I278: $\{\hat{\mathcal{Y}}_{\epsilon_M}^M(\omega), m \geq 1\}$

I279: $H(\mathcal{Y}) \oplus B(0, \bar{\epsilon})$
 I280: $y \in G \cap (H(\mathcal{Y}) \oplus B(0, \bar{\epsilon})) \neq \emptyset$
 I281: z_1^y, z_2^y
 I282: $\bar{\epsilon} > 0$
 I283: $G_\partial^1 = \bar{B}_\partial^1(z_1^y, \bar{\epsilon})$
 I284: $G_\partial^2 = \bar{B}_\partial^2(z_2^y, \bar{\epsilon})$
 I285: $G_\partial^1, G_\partial^2$
 I286: $\bar{\epsilon}$
 I287: $\hat{\mathcal{Y}}_{\bar{\epsilon}-\epsilon}^M$
 I288: $x \in \mathbb{R}^p$
 I289: $Y \in \mathcal{K}$
 I290: $\partial\mathcal{Y} \subseteq Y \oplus B(0, \epsilon)$
 I291: $Y \subseteq \mathcal{Y}$
 I292: $d_H(H(Y), H(\mathcal{Y})) \leq \epsilon$
 I293: $\partial\mathcal{X} \subseteq \mathcal{X}$
 I294: $D(\partial\mathcal{X}, \epsilon)$
 I295: $\tilde{x} \in B(x, \epsilon)$
 I296: $\|\tilde{x} - x\| \leq \epsilon$
 I297: $\|f(\tilde{x}) - y\| \leq L \|\tilde{x} - x\| \leq L\epsilon \leq \delta$
 I298: $\epsilon \leq \delta/L$
 I299: $F_{\partial\mathcal{X}} = \{x_i\}_{i=1}^{|F_{\partial\mathcal{X}}|} \subseteq \partial\mathcal{X}$
 I300: (ϵ/L)
 I301: $|F_{\partial\mathcal{X}}| = D(\partial\mathcal{X}, \epsilon/L)$
 I302: $x_i \in F_{\partial\mathcal{X}}$
 I303: $\|x - x_i\| \leq \epsilon/L$
 I304: $F_{\partial\mathcal{Y}} = \{f(x_i) \mid x_i \in F_{\partial\mathcal{X}}\}$
 I305: $F_{\partial\mathcal{Y}}$
 I306: $|F_{\partial\mathcal{Y}}| = |F_{\partial\mathcal{X}}| = D(\partial\mathcal{X}, \epsilon/L)$
 I307: $D(\partial\mathcal{Y}, \epsilon) \leq |F_{\partial\mathcal{Y}}| = D(\partial\mathcal{X}, \epsilon/L)$
 I308: $f(B(x, \epsilon)) \subset B(y, \delta)$
 I309: $\epsilon \in [0, \delta/L]$
 I310: $B(x, \epsilon) \subseteq f^{-1}(B(y, \delta))$
 I311: $F_{\partial\mathcal{Y}} = \{y_i\}_{i=1}^{|F_{\partial\mathcal{Y}}|} \subseteq \partial\mathcal{Y}$
 I312: $|F_{\partial\mathcal{Y}}| = D(\partial\mathcal{Y}, \epsilon)$
 I313: $y_i \in F_{\partial\mathcal{Y}}$
 I314: $\|y - y_i\| \leq \epsilon$
 I315: $B(F_{\partial\mathcal{Y}}, \epsilon) = \bigcup_{y_i \in F_{\partial\mathcal{Y}}} B(y_i, \epsilon)$
 I316: $\partial\mathcal{Y} \subset B(F_{\partial\mathcal{Y}}, \epsilon)$
 I317: $\partial\mathcal{Y} \not\subseteq Y_{2\epsilon}^M \implies B(F_{\partial\mathcal{Y}}, \epsilon) \not\subseteq Y_{2\epsilon}^M$
 I318: $H(\mathcal{Y}) = H(\partial\mathcal{Y}) \subseteq H(Y \oplus B(0, \epsilon)) = H(Y) \oplus B(0, \epsilon)$
 I319: $H(Y) \subseteq H(\mathcal{Y}) \subset H(\mathcal{Y}) \oplus B(0, \epsilon)$
 I320: $H(\mathcal{Y}) \subset H(Y) \oplus B(0, \epsilon)$
 I321: $H(Y) \subset H(\mathcal{Y}) \oplus B(0, \epsilon)$
 I322: $\delta_M = D(\partial\mathcal{X}, \epsilon/(2L)) (1 - \Lambda_\epsilon^L)^M$
 I323: $\partial\mathcal{Y} \not\subseteq f(\partial\mathcal{X})$
 I324: $\delta_M = D(\mathcal{X}, \epsilon/(2L))(1 - \Lambda_\epsilon^L)^M$
 I325: $Y_\epsilon^M = \bigcup_{i=1}^M B(y_i, \epsilon)$
 I326: $Y^M = \{y_i\}_{i=1}^M$
 I327: y
 I328: $\pi(\partial\mathcal{Y}, Y_\epsilon^M)$
 I329: $\mathbb{P}_\mathcal{Y}(B(y, \epsilon)) \geq \mathbb{P}_\mathcal{X}(B(x, \epsilon/L))$
 I330: $D(\partial\mathcal{Y}, \epsilon/2) \leq D(\partial\mathcal{X}, \epsilon/(2L))$
 I331: $\partial\mathcal{Y} \subseteq Y_\epsilon^M$
 I332: $Y^M \subseteq \mathcal{Y}$
 I333: 8
 I334: $d_H(H(Y^M), H(\mathcal{Y})) \leq \epsilon$

$$\begin{aligned}
\text{I335: } & \hat{\mathcal{Y}}^M = \mathcal{H}(Y^M) \\
\text{I336: } & d_H(\hat{\mathcal{Y}}^M, \mathcal{H}(\mathcal{Y})) \leq \epsilon \\
\text{I337: } & \mathcal{Y} \subseteq \mathcal{H}(\mathcal{Y}) \subseteq \hat{\mathcal{Y}}^M \oplus B(0, \epsilon) = \hat{\mathcal{Y}}_\epsilon^M \\
\text{I338: } & \mathbb{P}_{\mathcal{Y}}(\mathcal{Y}) = 1 \\
\text{I339: } & \tilde{x} \in \text{Int}(\mathcal{X}) \\
\text{I340: } & B(\tilde{x}, r) \\
\text{I341: } & x \in B(\tilde{x}, r) \\
\text{I342: } & B(\tilde{x}, r) \subseteq \mathcal{X} \\
\text{I343: } & \|x - \tilde{x}\| = r \\
\text{I344: } & \inf_{x \in \partial \mathcal{X}} \lambda(\mathcal{X} \cap B(x, \epsilon)) \geq \lambda(B(0, \epsilon) \cap B(\mathbf{r}, r)) \\
\text{I345: } & \delta_M = D(\partial \mathcal{X}, \epsilon/(2L)) (1 - p_0 \Lambda_\epsilon^{r,L})^M \\
\text{I346: } & \mathbb{P}_{\mathcal{X}}(B(x, \epsilon/(2L))) \geq \Lambda_\epsilon^L \\
\text{I347: } & A \in \mathcal{B}(\mathcal{X}) \\
\text{I348: } & \mathbb{P}_{\mathcal{X}}(B(x, \epsilon/(2L))) = \mathbb{P}_{\mathcal{X}}(\mathcal{X} \cap B(x, \epsilon/(2L))) \\
\text{I349: } & \mathbb{P}_{\mathcal{X}}(B(x, \epsilon/(2L))) = \mathbb{P}_{\mathcal{X}}(\mathcal{X} \cap B(x, \epsilon/(2L))) \geq p_0 \lambda(\mathcal{X} \cap B(x, \epsilon/(2L))) \\
\text{I350: } & \mathbb{P}_{\mathcal{X}}(B(x, \epsilon/(2L))) \geq p_0 \lambda(B(0, \epsilon/(2L)) \cap B(\mathbf{r}, r)) \\
\text{I351: } & a \in \mathbb{R} \\
\text{I352: } & I_x(a, b) = \Gamma(a+b) \Gamma(a) \Gamma(b)^{-1} \int_0^x t^{a-1} (1-t)^{b-1} dt \\
\text{I353: } & c_1 = \frac{(\epsilon/(2L))^2}{2r} \\
\text{I354: } & c_1 = \frac{2r^2 - (\epsilon/(2L))^2}{2r} \\
\text{I355: } & \Lambda_\epsilon^{r,L} = V(\epsilon/(2L), c_1) + V(r, c_2) \\
\text{I356: } & f: \mathbb{R}^n \rightarrow \mathbb{R}^n \\
\text{I357: } & \ell \in \mathbb{N} \\
\text{I358: } & (W^k, b^k)_{k=0}^\ell \\
\text{I359: } & \phi^k: \mathbb{R}^{n_k} \rightarrow \mathbb{R}^{n_{k+1}} \\
\text{I360: } & \phi^k(x) = (\varphi(x_1), \dots, \varphi(x_{n_k})) \\
\text{I361: } & \varphi(z) = \max(0, z) \\
\text{I362: } & \mathcal{X} \subset \mathbb{R}^n \\
\text{I363: } & \mathcal{A} = \{A_1, \dots, A_N\} \\
\text{I364: } & A_i \subseteq \mathcal{X} \\
\text{I365: } & f|_{A_i} \\
\text{I366: } & \Lambda_N = \frac{\min_{i=1, \dots, N} \lambda(A_i)}{\lambda(\mathcal{X})} \\
\text{I367: } & L_i = \|\nabla f(x_i)\| \\
\text{I368: } & i = 1, \dots, M \\
\text{I369: } & \hat{L} = \max_{i=1, \dots, M} L_i \\
\text{I370: } & \hat{L} \\
\text{I371: } & \delta_M = N(1 - \Lambda_N)^M \\
\text{I372: } & L = \max_{i=1, \dots, N} \{\|\nabla f(x_i)\| \text{ for some } x_i \in A_i\} \\
\text{I373: } & x_j \\
\text{I374: } & \mathbb{P}(x_j \notin A_i) = 1 - \mathbb{P}(x_j \in A_i) \leq 1 - \Lambda_N \\
\text{I375: } & N \\
\text{I376: } & u \sim \text{Unif}(0, 1) \\
\text{I377: } & r \leftarrow u^{1/p} \\
\text{I378: } & z \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_p) \\
\text{I379: } & \mathbf{I}_p \in \mathbb{R}^{p \times p} \\
\text{I380: } & x \leftarrow (r \cdot z) / \|z\|_2 \\
\text{I381: } & \beta \\
\text{I382: } & u \sim \text{Beta}(\alpha, \beta) \\
\text{I383: } & \beta = 1 \\
\text{I384: } & \text{Beta}(1, 1) = \text{Unif}(0, 1) \\
\text{I385: } & D(\partial \mathcal{X}, \bar{\epsilon}) \leq (2\pi)/(2\bar{\epsilon}) + 1 \\
\text{I386: } & p_0^\alpha > 0 \\
\text{I387: } & \mathbb{P}_{\mathcal{X}}^\alpha(B_x) \geq p_0^\alpha \lambda(B_x) \\
\text{I388: } & B_x = B(x, \bar{\epsilon}) \cap \mathcal{X} \\
\text{I389: } & p_0^\alpha = (1 - \text{CDF}_{\text{Beta}(\alpha, \beta)}((1 - \bar{\epsilon})^p)) / (1 - \text{CDF}_{\text{Unif}(0, 1)}((1 - \bar{\epsilon})^p))
\end{aligned}$$

I390: $x_t \in \mathbb{R}^n$
 I391: $u_t \in \mathbb{R}^m$
 I392: $\mathcal{X}_0 \subset \mathbb{R}^n$
 I393: $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
 I394: $B = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$
 I395: $\mathcal{X}_0 = \{(x^1, x^2) \in \mathbb{R}^2 : 5 \leq 2x^1 \leq 6, -1 \leq 4x^2 \leq 1\}$
 I396: $K(x_1, x_2) = \exp(-\|x_1 - x_2\|/0.05)$
 I397: $M = 1400$
 I398: $\mathcal{X} = [2.5, 3] \times [-0.25, 0.25]$
 I399: $D(\partial\mathcal{X}, \epsilon/(2L)) \leq 2 \cdot (0.5 + 0.5)/(2\epsilon/(2L)) + 1 = 2/\epsilon + 1$
 I400: Λ_ϵ^L
 I401: $\Lambda_\epsilon^L = (2 \cdot (\epsilon/2L))/(4 \cdot 0.5) = \epsilon/2$
 I402: $M \geq \frac{\log(\delta_M) - \log(D(\partial\mathcal{X}, \epsilon/(2L)))}{\log(1 - \Lambda_\epsilon^L)} \approx 1376$
 I403: $\mathbf{r}$
 I404: $H(\cdot)$
 I405: $B(x, r)$
 I406: $D(A, d)$
 I407: $\hat{\mathcal{Y}}_\epsilon^M = \hat{\mathcal{Y}}^M \oplus B(0, \epsilon)$
 I408: Y
 I409: Y^M
 I410: Y_ϵ^M
 I411: d_H
 I412: d_H
 I413: n
 I414: $\mathcal{Y}_\epsilon^M$
 I415: $\{\mathcal{Y}^M\}$
 I416: $d_H(\mathcal{Y}^M, \mathcal{Y})$
 I417: $(2d\sqrt{n}/\epsilon)^n$
 I418: n
 I419: $d_H(\hat{\mathcal{Y}}^M, \mathcal{Y})$
 I420: H
 I421: $[-0.015, 0.015]$
 I422: $\mathcal{X}_t(\nu)$
 I423: G_∂^1
 I424: $\bigcap_{N=0}^\infty A_n \subseteq A_0$
 I425: $\epsilon \rightarrow \bar{\epsilon}$
 I426: G_δ^1, G_δ^2
 I427: $m \geq 1$
 I428: c_2
 I429: c_1
 I430: $V(r, a)$
 I431: p_0^α